PORTFOLIO · PRODUCTION AI ENGINEERING

# An AI agent you can defend in an audit.

**Trust-Native ReAct Agent Framework** — a full-stack agent platform where layer boundaries are enforced by CI, every authorization flows through a signed trust kernel, and every decision is replayable.

| | | |
|---|---|---|
| **108K+** | lines of code, Python + TypeScript | |
| **2,500** | test functions in 186 test files | |
| **8 wks** | first commit to deployable, solo | |

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THE PROBLEM IT SOLVES

# Most agent demos die in security review

Typical agent frameworks ship a clever loop and leave trust, auditability, and security as an exercise for the reader. This codebase inverts that: safety is a property of the architecture, not a TODO.

1

## Architecture with teeth

A four-layer design where dependencies flow downward only. 12 dedicated architecture test modules mechanically verify the boundaries — an illegal import fails the build.

2

## A signed trust kernel

Authorization decisions flow through a pure, dependency-free kernel (stdlib + Pydantic only). Capability and policy records are cryptographically signed; changes force re-signing.

3

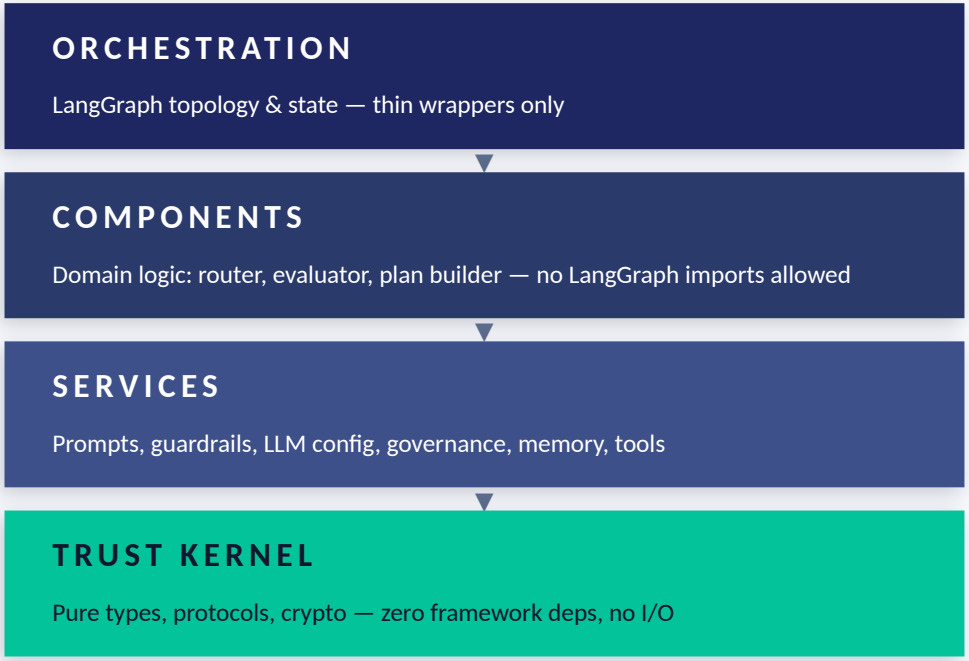
## A replayable audit trail

Black-box recordings, phase logs, and a signed agent-facts registry capture every LLM call, routing decision, and guardrail verdict — so “why did the agent do that?” has an answer.

Verified by static audit of github.com/rajnishkhatri/AgentsFramework · Jun 11, 2026

ARCHITECTURE

# Four layers, enforced by the test suite




*dependencies flow downward only*

## NOT A CONVENTION. A FAILING BUILD.

**The tests/architecture/ suite** contains 12 boundary-gate modules: dependency rules, service isolation, layer placement, read-only surfaces.

A component that imports the wrong layer breaks CI — the architecture cannot silently rot as the codebase grows.

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ENGINEERING DISCIPLINE

# A 2:1 test-to-code ratio, by design

**2,500**

**test functions**

across 186 files, organized by architectural layer

**53.6K**

**lines of test code**

vs 26.7K lines of Python application code

**0**

**live LLM calls in CI**

model calls mocked by marker scope; live evals run out-of-band

**12**

**architecture gates**

boundary tests run as their own CI job on Python 3.13

## HOW IT'S BUILT

Property-based testing (Hypothesis) on the trust kernel · rejection tests written before acceptance tests for every guard · hash-pinned **requirements.lock** with a CI job proving a cold pinned install stays green · every prompt externalized as a Jinja2 template (23 templates, zero prompt strings in Python)

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SECURITY & GOVERNANCE

# Defense in depth, then prove it happened

1. **Input guardrail**

   LLM-as-judge rejects prompt injection & system-prompt overrides

2. **Tool validators**

   Deterministic Pydantic gates: shell command allowlist, path sandboxing

3. **Output guardrail**

   Scans for PII, API keys, and system-prompt leakage (regex + LLM)

4. **Injection classifier**

   Fourth line of defense in the governance layer

## EVERY DECISION, REPLAYABLE

**Black-box recordings**
full input/output capture per run

**Phase logs**
structured trace of each reasoning phase

**Signed agent-facts registry**
capabilities & policies, GCS-backed in prod

**Explainability dashboard**
6 read-only modules: traces, decisions, guardrails, agents, compliance, live logs

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FULL-STACK & INFRASTRUCTURE

# Not a notebook — a deployable product

## FRONTEND

### 27.7K lines of TypeScript

- Next.js 15 · React 19 · CopilotKit UI
- WorkOS auth · Zod · Tailwind v4 / shadcn
- 75 frontend test files + read-only explainability app

## RUNTIME & MIDDLEWARE

### 26.7K lines of Python

- LangGraph orchestration · LiteLLM multi-provider routing
- Ports-and-adapters middleware: WorkOS JWT, Mem0 memory, Langfuse telemetry
- Postgres checkpointing · OpenAPI contract

## INFRA & DELIVERY

### 20 Terraform files, 10 Rego policies

- Cloud Run + Cloudflare edge + Neon Postgres dev tier
- Docker images for agent, backend, search
- 4 GitHub Actions workflows incl. wire-format baselines

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WHAT THIS CODEBASE DEMONSTRATES

# Architecture leadership you can diff.

### Systems thinking

Four-layer design with mechanical enforcement — the kind of governance regulated AI teams need on day one.

### Test-first discipline

2:1 test-to-code ratio, property-based testing, failure-path-first guards.

### Regulated-AI fluency

Signed audit trails, explainability UI, OPA policies — built for the "explain this decision" conversation.

### Velocity, solo

146 commits, Apr 17 → Jun 10, 2026: runtime, frontend, middleware, and infra by a single engineer.

**Read the fine print (on purpose):** single-maintainer project, license pending, pre-1.0. Live-LLM evals run outside CI by design. Every figure here comes from a static audit of the repository — nothing is projected or extrapolated.

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